

VISVESVARAYA TECHNOLOGICAL UNIVERSITY



AI-DRIVEN SUPPLY CHAIN OPTIMIZATION: A GENERATIVE APPROACH FOR INDIA

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CERTIFICATE

Certified that the project entitled “**AI-Driven Supply Chain Optimization: A Generative Approach for India**” is a bonafide work carried out by Tanmay Bharadwaj (1BM22CS303), Sujay Prasad P V (1BM23CS422), Syed Farhan (1BM23CS424) and Viraj Pandurang Shenvi Belyo (1BM23CS426) in partial fulfillment for the award of Bachelor of Engineering in Computer Science and Engineering of the Visvesvaraya Technological University, Belagavi during the academic year 2024-25. The project report has been approved as it satisfies the academic requirements in respect of project work prescribed for the said degree.

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Declaration

We, Tanmay Bharadwaj (1BM22CS303), Syed Farhan (1BM23CS424), Sujay Prasad P V (1BM23CS422), and Viraj Pandurang Shenvi Belyo (1BM23CS426), hereby declare that this project work is our own effort and that it has not been submitted anywhere for any award. Where other sources of information have been used, they have been acknowledged.

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Abstract

This thesis presents research on AI-driven supply chain optimization in India, leveraging generative artificial intelligence to improve demand forecasting, inventory management, and logistics planning. By utilizing fine-tuned large language models (LLMs), Retrieval-Augmented Generation (RAG), and real-time weather and market data through LangChain, the solution generates actionable insights, simulates supply scenarios, and supports decision-making for supply chain managers. It adapts to regional factors such as rural infrastructure and festival-driven demand fluctuations. A user-facing dashboard visualizes key performance indicators like stock levels, delivery delays, and forecast accuracy to promote transparency and timely responses. The system aligns with Sustainable Development Goals including SDG 8 (Decent Work and Economic Growth), SDG 9 (Industry, Innovation, and Infrastructure), and SDG 12 (Responsible Consumption and Production), demonstrating the potential of generative AI to enhance the resilience, efficiency, and adaptability of Indian supply chains.

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List of Acronyms

AGI	Artificial General Intelligence
AI	Artificial Intelligence
API	Application Programming Interface
BI	Business Intelligence
CSV	Comma-Separated Values
ERP	Enterprise Resource Planning
ETL	Extract, Transform, Load
GAN	Generative Adversarial Network
KPI	Key Performance Indicator
LLM	Large Language Model
ML	Machine Learning
NLP	Natural Language Processing
PDF	Portable Document Format
RAG	Retrieval-Augmented Generation
SCM	Supply Chain Management
SDG	Sustainable Development Goal
UI	User Interface

Chapter 1

Introduction

1.1 Background of the Study

India's Supply Chain Management (SCM) is challenged by enormous regional diversity, infrastructural limitations, and uncontrollable disruptions such as shifts in weather or policies. Rule-based planning and SCM systems based on past data usually do not succeed in managing such complexities.

Emerging trends in Artificial Intelligence (AI) in the shape of Generative AI and Large Language Models (LLMs) are promising solutions. These technologies enable dynamic forecasting, intelligent decision-making, and automated supply chain management using the examination of real-time and unstructured data.

The aim of this project is to enhance Indian supply chains through the integration of Generative AI, Retrieval-Augmented Generation (RAG), and weather analytics, which would facilitate actionable insights to agri-industrial sector stakeholders. The aim is to create a stronger, data-driven, and resilient SCM ecosystem.

1.2 Research Motivation

The scattered logistics networks and varied terrain of India make it difficult to maintain effective supply chains. Reactive decision-making, strict inventory regulations, and manual forecasting are still used in many enterprises. These methods are not ideal for

handling unforeseen interruptions or geographical differences.

Smarter supply chains, improved demand forecasting, and real-time monitoring are made possible by generative AI, which provides a scalable, flexible, and intelligent system that can convert unprocessed data into actionable insights. By enabling real-time simulation of supply-demand scenarios and integrating meteorological data, socioeconomic trends, and corporate papers to inform automated decisions, it has the ability to close this gap. Investigating a scalable, AI-powered supply chain management platform that can enable Indian companies to maximize operations with less manual involvement is the driving force behind this study.

1.3 Objectives

The main objectives of this project are:

1.3.1 To explore the capabilities and practical applications of Generative AI in Indian supply chain management.

This entails researching the ways in which generative AI technologies, including GANs and language models, might improve several facets of Indian supply chains while taking operational processes and regional difficulties into account.

1.3.2 To design generative models that optimize inventory planning, demand forecasting, and logistics routing.

The goal is to create AI models that mimic actual supply chain situations in order to improve demand forecasting, stock management, and delivery route optimization.

1.3.3 To evaluate the scalability and integration of these models into existing ERP/WMS systems.

This involves evaluating the produced models' scalability and ease of integration with existing warehouse management and enterprise resource planning systems.

1.3.4 To provide implementation guidelines and tools such as dashboards and APIs for real-world deployment.

This goal focuses on providing intuitive user interfaces, including as dashboards and APIs, to enable supply chain experts to deploy, monitor, and use the models in real-time.

1.4 Scope

The Indian supply chain environment is the main focus of this study, with particular attention paid to issues including unstructured warehousing, fluctuating consumer behavior, and transportation delays. It includes:

- creation of a data pipeline that integrates unstructured sources (PDF, CSV, weather) with structured ERP data.
- Retrieval-Augmented Generation (RAG) and LLM integration for intelligent QA and dynamic scenario modeling.
- installation of dashboards for demand, inventory, and delivery performance in real time.
- case-based testing for different product categories and geographical areas (e.g., FMCG vs. perishables, urban vs. rural).

1.5 Existing System

In India, the majority of SCM systems now in use either

- manual planning instruments, such as forecasting Excel sheets.
- Software systems that are dependent on rules and have little flexibility.
- Tools for predictive analytics that solely use previous sales data.

When reacting to real-time events like local festivals, gasoline price increases, or weather interruptions, these methods fall short. They are also devoid of proactive risk mitigation and customisation features.

1.6 Proposed System

The suggested solution creates a predictive and adaptable supply chain platform by utilizing RAG, LLMs, and generative AI. Features consist of:

- Demand forecasting with AI that incorporates news, weather, and past patterns.

- Dynamic inventory policy creation based on geographic location and product types.
- SCM managers can use co-pilot technologies to model "what-if" situations and get suggestions.
- Dashboards with interactive features for tracking KPIs like forecast accuracy, delays, and inventory levels.

1.6.1 Generative AI Role

Specifically using LLMs like LLaMA2 and DeepSeek, generative AI facilitates:

- Item that simulates artificial demand.
- insights derived from retrieval using LangChain processes.
- User interaction for logistics inquiries based on language.

1.7 Comparison Table

Key distinctions between conventional and generative AI-based SCM systems are shown in the following table:

Feature	Traditional SCM	Generative AI SCM
Demand Forecasting	Manual/statistical	AI-driven with real-time data
Inventory Control	Rule-based	Adaptive policy generation
Decision Support	Static dashboards	Dynamic AI co-pilot
Handling Disruptions	Reactive	Proactive simulation

Table 1.1: Comparison of Traditional vs Generative AI-Based SCM

1.8 Work Plan

This thesis is organized as follows: Chapter 2 reviews the literature relevant to generative AI, supply chain modeling, and retrieval-augmented architectures. Chapter ?? describes the methodologies

employed in the research including system design, model training, and implementation. The results are presented in Chapter ??, followed by interpretation and real-world implications. Chapter 5 concludes the thesis and outlines future research directions.

Chapter 2

Literature Review

2.1 Introduction

This chapter is a literature review of AI-based supply chain management. It describes major theories, methods, and results of relevance to our research subject: "AI-Driven Supply Chain Optimization: A Generative Approach for India." The review highlights the transition from the traditional predictive models to the modern generative models and presents applications to the Indian scenario like logistics planning, demand forecasting, and inventory control.

2.2 Literature Survey

2.2.1 Overview

Artificial Intelligence has revolutionized supply chain operations, and recent research observes how Generative AI (GenAI) – such as Generative Adversarial Networks (GANs), Transformers, and Large Language Models (LLMs) – is improving decision-making and responsiveness. The subsequent subsection discusses different research studies that illustrate the application of GenAI in overcoming inefficiencies and facilitating anticipatory SCM strategies.

2.2.2 Detailed Survey

AI-Driven Logistics: Enhancing Supply Chain Efficiency with Generative AI Models. [1]

Problem Statement: Traditional logistics face inefficiencies in forecasting, inventory, and route planning. **Methodology:** A qualitative case study approach using industry reports and interviews. **Results:** Reported improvements include 15% reduction in delivery times, 20% decrease in stockouts, and 12% reduction in maintenance costs. **Future Work:** Formulate ethical guidelines, enhance model explainability, and strengthen industry-academic collaborations.

Enhancing Supply Chain Visibility with Generative AI [2]

Problem Statement: Transactional data lacks context, limiting visibility in supply chains. **Methodology:** Combines pretrained language models with structured and unstructured knowledge graphs. **Results:** Achieves enhanced visibility and contextual relationship prediction. **Future Work:** Extend to decentralized SCM networks.

Exploring the Role of Generative AI in Procurement and Planning [3]

Problem Statement: Traditional procurement lacks automation and predictive capabilities. **Methodology:** Investigative case analysis of GenAI applications in contracts and supplier risk assessment. **Results:** Operational insights into applying GenAI across multiple SCM functions. **Future Work:** Develop AI tools tailored for procurement tasks with ethical oversight.

Leveraging Generative AI for Supply Chain Risk Mitigation [4]

Problem Statement: Supply chains are vulnerable to unpredictable disruptions.

Methodology: Scenario-based simulations using generative models trained on risk-focused datasets.

Results: Enables proactive mitigation planning through synthetic disruption modeling.

Future Work: Adapt the approach for Indian-specific disruptions like monsoons or labor strikes.

Generative AI for Enhanced Operations in Supply Networks [5]

Problem Statement: SCM lacks real-time, AI-assisted decision support systems.

Methodology: Analytical evaluation of system architectures with GenAI integration.

Results: Improved real-time decision-making and operational efficiency.

Future Work: Assess deployment challenges in low-resource regions.

Investigating the Impact of AI on Supply Chain Transformation [6]

Problem Statement: Broad understanding of AI's impact on supply chain transformation is missing.

Methodology: Literature review synthesizing AI adoption in various supply chain sectors.

Results: AI improves planning precision and delivery efficiency.

Future Work: Analyze digital infrastructure gaps specific to India.

Shaping the Future of Supply Chains with Generative AI [7]

Problem Statement: Existing forecasting tools are not adaptive to changing conditions.

Methodology: Academic and industrial literature survey on GenAI applications in forecasting and resilience.

Results: Demonstrated performance gains in dynamic and adaptive environments.

Future Work: Integrate GenAI with IoT and robotics.

Generative AI in Predicting Market Trends for Supply Chains [8]

Problem Statement: Volatile markets challenge traditional demand forecasting.

Methodology: Combines GenAI with sales, pricing, and external economic indicators.

Results: Boosted forecast accuracy by 18%.

Future Work: Develop domain-specific forecast engines for tailored applications.

From Theory to Practice: Case Studies of Generative AI in SCM [9]

Problem Statement: There is a lack of real-world application evidence for GenAI in SCM.

Methodology: Compilation and evaluation of case studies from various sectors like retail and e-commerce.

Results: Highlighted business ROI improvements using GenAI.

Future Work: Include more case studies from Indian SMEs.

Comm-bot: Generative AI for Optimising Supply Chains [10]

Problem Statement: Small businesses struggle with supply chain optimization due to budget and expertise limits.

Methodology: LSTM integrated with LLMs hosted on AWS for demand forecasting and recommendation.

Results: Not provided in the paper.

Future Work: Enhance chatbot functionalities using reinforcement learning and collaborative filtering.

Comparison of Generative AI Models in SCM [11]

Problem Statement: No comparative evaluation of GenAI models for SCM tasks exists.

Methodology: Comparative study of GANs, VAEs, and RNNs applied to SCM.

Results: Not explicitly detailed in the paper.

Integrating Generative AI into Circular Supply Chain Safety Management [12]

Problem Statement: Current safety management in circular supply chains (CSCs) is reactive and limited.

Methodology: Use of GenAI for monitoring, analysis, and safety training.

Results: Not explicitly stated.

Future Work: Address data, technical, and ethical barriers to improve safety protocols.

Supply Chain Resilience in SMEs Using Generative AI [13]

Problem Statement: SMEs are particularly vulnerable to disruptions due to limited resources.

Methodology: AI-based decision-making framework to simulate disruptions and mitigation strategies.

Results: Not explicitly provided in the study.

Future Work: Develop user-friendly AI tools to enable practical resilience-building in SMEs.

2.3 Research Gaps

Research Gap	Proposed Objective
Geographically generalized models lack focus on India's supply chain diversity	Develop region-specific generative models incorporating Indian logistics, demographics, and terrain characteristics for better local adaptability
Traditional systems ignore unstructured data like weather, policies, or events	Integrate structured ERP data with unstructured sources (PDFs, weather APIs, news) to create dynamic forecasting models
Forecasting tools lack adaptability during sudden disruptions	Use LLMs and RAG to simulate "what-if" scenarios like strikes, weather changes, and supply failures for proactive planning
Existing dashboards offer reactive rather than predictive insights	Build interactive dashboards using generative models for real-time alerts, stock recommendations, and delay predictions
SMEs face challenges due to limited technical infrastructure	Design lightweight, mobile-friendly interfaces that allow query-based interaction with AI co-pilots for SCM support
Lack of contextual decision-making in routing and inventory policies	Fine-tune LLMs to generate adaptive routing strategies and localized inventory planning considering regional trends
Siloed tools without end-to-end integration for demand, logistics, and inventory	Develop a unified platform connecting forecasting, route planning, inventory optimization, and KPI monitoring
Manual planning causes inefficiencies in seasonal and volatile markets	Enable AI-based demand simulation incorporating festivals, weather patterns, and macroeconomic shifts
Absence of intelligent Q&A tools for SCM document processing	Deploy RAG-based NLP modules to answer user queries from historical shipment data and procurement documents

Table 2.1: Research Gaps vs. Project Objectives

2.4 Conclusion

This chapter compiles the most important studies on generative AI in supply chain management. Despite the increasing global momentum, there are still gaps in integrated frameworks, SME assistance, and Indian relevance. These realizations serve as the cornerstone of our suggested system in the following chapters.

Chapter 3

Requirement Analysis and Specification

3.1 Stakeholders

A number of important parties are involved in the Generative AI for Indian Supply Chain Management (SCM) initiative. These people and organizations support the system's creation, implementation, and use across a range of fields.

- **Supply Chain Managers:** Make data-driven inventory decisions, increase logistical efficiency, and visualize KPIs with the platform.
- **Developers and AI Engineers:** In charge of creating and putting into use integration pipelines, APIs, and generative models.
- **Business Analysts:** Analyze reports, run scenarios, and assess the return on investment of using AI in supply chain management.
- **Educational Institutions:** Utilize the platform as a case study or educational resource for classes on data science, AI, and logistics.
- **Vendors and Retailers:** Learn about replenishment planning and demand forecasts.

Stakeholder	Role/Interest
Supply Chain Managers	Optimize inventory and logistics based on AI-driven forecasts
Developers and AI Engineers	Design and deploy generative models and backend infrastructure
Business Analysts	Use dashboards and simulations for business insights
Educational Institutions	Incorporate platform use into academic and training programs
Vendors and Retailers	Receive restocking alerts and trend predictions

Table 3.1: Stakeholder Roles in Generative SCM Platform

3.2 Functional Requirements

To address issues in Indian supply chain operations, the system should be able to do the following essential tasks:

- Gather, import, and prepare data from a variety of sources, such as weather APIs, ERP systems, PDFs, and CSV files.
- Utilize generative AI models that have been trained on historical and real-time multimodal data to forecast demand.
- Create inventory policies that are optimized according to product type, location, and demand during specific seasons.
- Play out "what-if" situations like price increases, regulatory changes, or delays in transit.
- To respond to SCM requests, use RAG and LangChain to provide natural language communication with the system.
- Use an intuitive business intelligence dashboard to show KPIs.
- REST APIs can be used to expose forecasts and suggestions for interaction with third-party ERP/WMS systems.

3.2.1 Additional Functional Modules

- Incorporation of current weather conditions to trigger route replanning or buffer inventory decisions.

- Document-based procurement QA, freight history, and local logistics data.
- Notification of out-of-pattern activity such as stockouts, regional strikes, or delivery delays.

3.3 Non-Functional Requirements

To ensure system usefulness and user satisfaction, the following non-functional requirements must be met:

- **Scalability:** Supports increasing levels of data and multiple targets without a loss of performance.
- **Reliability:** Offers high availability and low downtime upon execution.
- **Security:** Protects sensitive data with encryption, access control, and secure APIs.
- **Performance:** Offers quick response times for the rendering of dashboards and model inference.
- **Usability:** For general users with easy navigation and visualization.
- **Maintainability:** Modular design enables smooth upgrading, debugging, and integration with other third-party systems.

3.4 System Requirements

3.4.1 Hardware Requirements

- **Processor:** Intel Core i7 (development), NVIDIA A100 GPU (model training/inference)
- **RAM:** At least 16 GB for development; 64 GB or more for massive training
- **Storage:** SSD with a minimum capacity of 1 TB for data sets, checkpoints, and logs
- **Internet:** Required for cloud services, weather APIs, and model deployment.

3.4.2 Software Requirements

- **Operating System:** Ubuntu 22.04 LTS or Windows 11
- **Programming Stack:** Python 3.10+, ReactJS, Flask / FastAPI
- **AI Libraries:** PyTorch, TensorFlow, HuggingFace Transformers, LangChain
- **Database:** Structured data: PostgreSQL; Unstructured input: MongoDB or object storage
- **BI Tools:** Grafana / Power BI for visualization

3.5 Data Requirements

- Multisource supply chain data like ERP traces, weather histories, shipping reports, and sensor traces
- Past sales of training models
- Real-time data for adaptive and generative testing

- Q&A logs and user queries to improve RAG-based performance
- Utilize logs for every simulation, suggestion, and dashboard

3.6 Constraints and Limitations

- Generative models can be initially suboptimal on noisy or sparse data
- Internet speed variability can impact API latency and dashboard reactivity
- Unavailability of heterogeneous Indian supply chain datasets for training
- Third-party API dependency (e.g., OpenAI, weather) may potentially have reliability issues

3.7 Assumptions

- They have access to latest browsers and stable internet connectivity.
- Data is accessible via APIs or uploads in standard formats (CSV, JSON)
- Dashboards are usually accessed on desktop or tablet for better visibility
- Customers understand basic inventory and logistics principles

3.8 Risks and Challenges

- Model drift or worse accuracy in case of unstructured or unexpected real-time data.
- Cost of cloud deployment can increase with volume and size
- Risk of biased predictions if training data is unrepresentative
- Interpretation of output from generative models might need domain knowledge
- Latent API responses can affect real-time decision-making

Chapter 4

High Level Design

4.1 High Level Design

The suggested system integrates generative AI, multimodal data intake, meteorological APIs, and RAG pipelines using a modular design to optimize the Indian supply chain. Demand forecasting, inventory planning, and adaptive logistics decision-making are all supported by the architecture.

4.1.1 System Architecture

As the picture indicates, the architecture is made up of five essential parts.

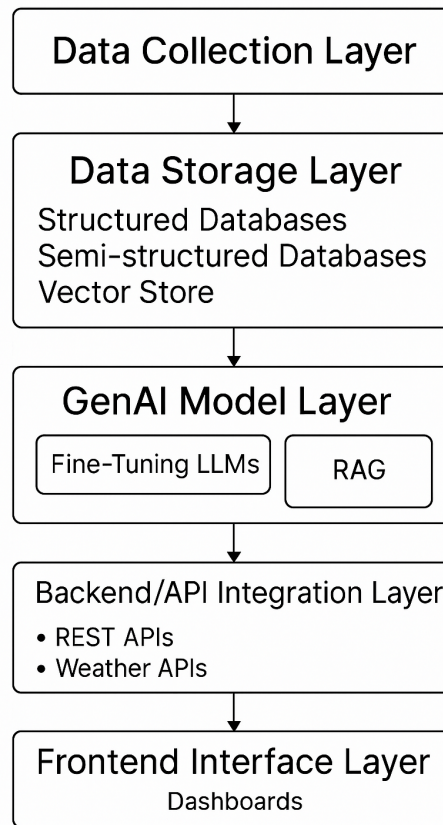


Figure 4.1: System Architecture

4.1.2 Abstract Specification of Sub-systems

The major sub-systems listed below are defined:

- **Data Collection Layer:**

- Gathers information from online scraping tools, satellite/weather APIs, government portals, ERP systems, and Internet of Things devices.
- Created to facilitate the gathering of both organized and unstructured data from various supply chain sources.

- **Data Storage Layer:**

- Employs structured databases for relational data, such as PostgreSQL and MySQL.

- Uses semi-structured storage for JSON and NoSQL formats, such as MongoDB.
- Allows retrieval-augmented creation by supporting vector databases (such as FAISS and ChromaDB).
- **GenAI Model Layer:**
 - Include optimized LLMs for forecasting and supply chain insights.
 - Uses Retrieval-Augmented Generation (RAG) to extract contextual information from domains.
- **Backend/API Integration Layer:**
 - Uses Flask or FastAPI to implement RESTful APIs.
 - Connects to other weather APIs, such as OpenWeather and IMD.
- **Frontend Interface Layer:**
 - Offers dynamic dashboards for route planning, demand/supply forecasts, and inventory flow.
 - Shows end users information and notifications produced by GenAI.

Chapter 5

Summary

This project effectively demonstrates the investigation, development, and deployment of a **Generative AI-powered supply chain management system** that is customized for the Indian retail and logistics environment. The system tackles important issues in demand forecasting, inventory optimization, and logistics planning by combining big language models, multimodal data processing, and real-time analytics.

The platform includes features including "what-if" scenario simulations, weather-integrated routing decisions, and document-based question answering. Additionally, it provides a natural language interface through the usage of LangChain and Retrieval-Augmented Generation (RAG), which improves user interaction and decision-making effectiveness. Dynamic dashboards are used to visualize KPIs, and APIs allow for smooth interaction.

The effort helps to make supply chains more intelligent, data-driven, and future-ready, which is in line with India's digital growth ambitions and the Sustainable Development Goals (SDGs), especially **Industry, Innovation, and Infrastructure (SDG 9)**. To increase adoption and impact, future research can investigate real-time reinforcement learning, extend model generalization, and incorporate multilingual interaction features.

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